

Longitudinal Dynamics and Control of a Fixed-Wing UAV

Trim, Linearisation, Modal Analysis and Autopilot Design

Sashank

AE695 — State Space Analysis and Design
Indian Institute of Technology Bombay

October 5, 2026

Abstract

This report implements the complete longitudinal design chain of Beard and McLain [1], chapters 3 to 6, for a fixed-wing unmanned aircraft: the nonlinear five-state plant, trim, linearisation about trim, transfer function models, the reduced-order short-period and phugoid approximations, and a successive-loop-closure autopilot. Everything is implemented in base MATLAB with no toolbox dependencies.

Two airframes are carried in parallel: the Aerosonde of the reference text, and a canard configuration taken from an existing six-degree-of-freedom Simulink model. Trim is solved to a residual of $\sim 2 \times 10^{-15}$ for both. The linearised model reproduces the published reference matrices to 3.8×10^{-5} (A_{lon}) and 1.4×10^{-10} (B_{lon}), and the analytic transfer function coefficients a_{θ_1} through a_{θ_3} match to seven significant figures.

Two defects in the reference material are documented. Its companion code contains no implemented physics, and its published trim point is not a force equilibrium: evaluating the dynamics there gives $\dot{u} = -0.85 \text{ m/s}^2$, because the propeller model delivers about 0.95 N against 10.3 N of drag. The verification strategy was restructured around both findings.

Contents

1	Scope and approach	3
1.1	State and input	3
2	Nonlinear longitudinal model (chapters 3 and 4)	3
2.1	Equations of motion	3
2.2	Aerodynamic forces and moments	4
2.3	Lift model	4
2.4	Drag model	4
2.5	Propulsion	4
3	Trim	5
3.1	The problem is triangular	5
3.2	Why not an optimiser	5
3.3	Results	6
4	Linearisation	6
4.1	Method	6
4.2	Results	6
4.3	Structural observations	7

5	Transfer function models	7
6	Reduced-order modes	7
7	Longitudinal autopilot	10
7.1	Structure	10
7.2	Altitude state machine	11
7.3	Implementation notes	11
7.4	Results	11
8	Verification	14
9	Defects in the reference material	15
9.1	The companion code contains no implemented physics	15
9.2	The published trim point is not a force equilibrium	15
9.3	The shipped parameter sets disagree with each other	15
10	Conclusions	16
A	Software	16

1 Scope and approach

The full rigid-body aircraft has twelve states. Chapter 5 of [1] decouples these into a lateral set (v, p, r, ϕ, ψ) and a longitudinal set (u, w, q, θ, h) , leaving north and east position as pure navigation states that appear in neither. This report takes the longitudinal set only.

The decoupling is obtained by forcing every lateral state to zero,

$$v = p = r = \phi = \psi = \beta = 0, \quad (1)$$

together with zero wind. Under (1) the Euler kinematics collapse from $\dot{\theta} = q \cos \phi - r \sin \phi$ to simply $\dot{\theta} = q$.

A consequence worth stating at the outset: the only attitude variable remaining is θ , which is an Euler angle with no singularity anywhere in the longitudinal flight envelope. The quaternion formulation that the reference text's companion code uses for its thirteen-state propagation, together with its quaternion-to-Euler conversion layer, is therefore unnecessary here and has been omitted. This removes the most error-prone part of the reference implementation without any loss of generality in the longitudinal plane.

1.1 State and input

$$x_{\text{lon}} = [u \quad w \quad q \quad \theta \quad h]^\top, \quad u_{\text{lon}} = [\delta_e \quad \delta_t]^\top, \quad (2)$$

with body-axis velocities in m/s, pitch rate in rad/s, pitch angle in rad, altitude in m, elevator in rad and throttle on $[0, 1]$.

Altitude is carried as a state rather than down-position p_d . This is a deliberate choice: the reference implementation integrates p_d and must then negate a row and a column of its Jacobian to report A_{lon} in the ordering of (2). Integrating h directly makes that post-processing step vanish, and the Jacobian emerges in the book's ordering and sign convention with no manipulation at all.

2 Nonlinear longitudinal model (chapters 3 and 4)

2.1 Equations of motion

Reducing book equations (5.3), (5.4), (5.6) and (5.11) under (1) gives

$$\dot{u} = -qw + f_x/m, \quad (3)$$

$$\dot{w} = qu + f_z/m, \quad (4)$$

$$\dot{q} = M_y/J_y, \quad (5)$$

$$\dot{\theta} = q, \quad (6)$$

$$\dot{h} = u \sin \theta - w \cos \theta. \quad (7)$$

The pitch equation (5) is a genuine simplification rather than an approximation: the general form carries $\Gamma_5 pr + \Gamma_6(p^2 - r^2)$, both of which vanish identically when $p = r = 0$.

Air data follows from the velocity components,

$$V_a = \sqrt{u^2 + w^2}, \quad \alpha = \text{atan2}(w, u). \quad (8)$$

The two-argument arctangent is used in place of $\arctan(w/u)$. The two agree for $u > 0$ but only the former stays finite at $u = 0$.

2.2 Aerodynamic forces and moments

With $\bar{q} = \frac{1}{2}\rho V_a^2$ and the non-dimensionalising length $\hat{c} = c/(2V_a)$,

$$C_L = C_L(\alpha) + C_{L_q} \hat{c} q + C_{L_{\delta_e}} \delta_e, \quad (9)$$

$$C_D = C_D(\alpha) + C_{D_q} \hat{c} q + C_{D_{\delta_e}} \delta_e, \quad (10)$$

$$C_m = C_{m_0} + C_{m_\alpha} \alpha + C_{m_q} \hat{c} q + C_{m_{\delta_e}} \delta_e. \quad (11)$$

Two remarks on (11). First, the correct non-dimensionalising length for the pitch-damping term is the *chord*. Using the span instead is an error present in some existing fixed-wing Simulink models, and it over-weights pitch damping by the ratio b/c , roughly a factor of five for the canard geometry considered here. Second, C_{m_0} in (11) is a zero-incidence derivative, not the constant term of a polynomial fit; the two meanings collide in some parameter files and a model can compile with the wrong one and still be silently wrong.

Lift and drag act in the wind frame and are rotated into body axes, after which gravity and thrust are added:

$$f_x = -mg \sin \theta + (-D \cos \alpha + L \sin \alpha) + T, \quad (12)$$

$$f_z = mg \cos \theta + (-D \sin \alpha - L \cos \alpha). \quad (13)$$

Thrust appears only in (12). That it is absent from (13) is not incidental; it is the structural fact that makes the trim computation of Section 3 triangular.

2.3 Lift model

For the Aerosonde, lift blends a linear model into a flat-plate model past stall through the sigmoid of book equations (4.9) and (4.10),

$$C_L(\alpha) = (1 - \sigma(\alpha))(C_{L_0} + C_{L_\alpha} \alpha) + \sigma(\alpha) 2 \operatorname{sign}(\alpha) \sin^2 \alpha \cos \alpha, \quad (14)$$

with σ the blending function parameterised by M and α_0 .

For the canard, α_0 , M and ϵ were never identified, so the lift curve is kept linear and the blend is disabled by a flag. No attempt was made to invent stall parameters. In particular, “no stall” must not be encoded as $\alpha_0 \rightarrow \infty$, because the sigmoid then evaluates to NaN.

2.4 Drag model

Drag is implemented as a polynomial in α evaluated by Horner’s method. This single code path covers both airframes, because the book’s linear-in- α drag is simply a degenerate quadratic:

$$C_D(\alpha) = \begin{cases} 0.132 \alpha + 0.0424 & \text{Aerosonde,} \\ 2.8252 \alpha^2 - 0.0137 \alpha + 0.0461 & \text{canard.} \end{cases} \quad (15)$$

It is worth recording that the Aerosonde drag is the linear form and *not* the induced-drag polar $C_{D_p} + C_L^2/(\pi e AR)$. This was established numerically rather than assumed: only the linear form reproduces the published a_{V_1} , as shown in Section 8. Consequently C_{D_p} , the aspect ratio and the Oswald efficiency are unused.

2.5 Propulsion

Two models are implemented, selected per airframe. The Aerosonde uses the propeller and motor model of the text’s addendum: the propeller torque is matched to the motor torque,

which gives a quadratic in propeller speed Ω ,

$$\underbrace{\frac{\rho D^5 C_{Q_0}}{4\pi^2}}_a \Omega^2 + \underbrace{\left(\frac{\rho D^4 C_{Q_1} V_a}{2\pi} + \frac{K_Q^2}{R} \right)}_b \Omega + \underbrace{\rho D^3 C_{Q_2} V_a^2 - \frac{K_Q V_{in}}{R} + K_Q i_0}_c = 0, \quad (16)$$

after which thrust follows from the thrust coefficient at the resulting advance ratio. Equation (16) is solved as $\Omega = -2c/(b + \sqrt{b^2 - 4ac})$ rather than by the textbook quadratic formula. The two are algebraically identical, but at cruise $4ac \ll b^2$, and the textbook form then loses most of its significant figures to cancellation.

The canard uses a velocity-lapsed law, $T = \max(0, T_{\text{static}}(1 - V_a/V_{\text{max}})\delta_t)$.

The legacy simple propeller model of chapter 4 is *not* implemented. It cannot be what generated the published reference values: it yields $\partial T/\partial V_a = -6.43 \text{ N}/(\text{m/s})$ where -2.35 is required.

3 Trim

3.1 The problem is triangular

Appendix F.2 of [1] shows that for wings-level longitudinal flight the trim problem decouples completely. Given a target airspeed V_a^* and flight path angle γ^* :

1. α fixes the kinematics: $\theta = \alpha + \gamma$, $q = 0$, $u = V_a \cos \alpha$, $w = V_a \sin \alpha$.
2. δ_e is then available in *closed form* from $\dot{q} = 0$ with $q = 0$, using (5) and (11):

$$\delta_e^* = -\frac{C_{m_0} + C_{m_\alpha} \alpha^*}{C_{m_{\delta_e}}}. \quad (17)$$

3. α follows from a *scalar* root-find on $\dot{w} = 0$. Because thrust is absent from (13), this step is independent of throttle.
4. δ_t follows from a scalar root-find on $\dot{u} = 0$, solved last.

Altitude needs no equation. Once $\theta = \alpha + \gamma$, substituting into (7) gives

$$\dot{h} = V_a \cos \alpha \sin(\alpha + \gamma) - V_a \sin \alpha \cos(\alpha + \gamma) = V_a \sin \gamma \quad (18)$$

identically, so the commanded climb rate is satisfied automatically. This is used as a verification condition in Section 8.

3.2 Why not an optimiser

The reference implementation solves trim with `fmincon` on a seventeen-element decision vector, because it trims all twelve states simultaneously and must enforce quaternion normalisation as an explicit constraint. For the longitudinal plane that is a constrained nonlinear program applied to a problem that is triangular and has an analytic middle step.

Two scalar `fzero` calls are used instead. The practical consequences are: no Optimization Toolbox dependency; a residual at machine precision rather than at a tolerance one has to choose; and an informative failure mode, since the bracket $\alpha \in [-0.2, 0.3]$ rad is physically meaningful and a sign-change failure genuinely means no trim exists at that airspeed. The last point matters downstream, because the Jacobian of Section 4 is taken about the trim point and any trim error propagates directly into it.

Quantity	Aerosonde	Canard
V_a^* [m/s]	25.00	25.00
α^* [deg]	2.8476	-0.3817
θ^* [deg]	2.8476	-0.3817
u^* [m/s]	24.9691	24.9994
w^* [m/s]	1.2420	-0.1666
δ_e^* [deg]	-7.1000	0.6389
δ_t^*	0.7734	0.1533
$\ f(x^*, u^*)\ $	1.91×10^{-15}	2.96×10^{-15}

Table 1: Trim at $V_a^* = 25$ m/s, $\gamma^* = 0$. The residual is $\|f(x^*, u^*)\|$ over the first four states, i.e. how nearly the computed point is a true equilibrium.

3.3 Results

Both airframes trim to machine precision. The canard trims at a slightly *negative* angle of attack, which is correct for it: its zero-incidence lift coefficient $C_{L_0} = 0.3494$ slightly exceeds the $C_L = 0.324$ needed to carry 2.4 kg at 25 m/s on 0.19 m^2 , so a small negative incidence is required to shed the excess.

4 Linearisation

4.1 Method

The Jacobians are formed by finite differences about the trim point, following Appendix F.4. Central differences with a per-variable scaled step $h_i = 10^{-6} \max(1, |x_i|)$ are used by default, giving second-order accuracy.

A forward-difference mode with a fixed step of 0.01 is also provided, used only for the reference comparison in Section 8: the published matrices were generated that way, and reproducing them requires reproducing their differencing scheme. That step is far too large for accuracy and is deliberately not the default. Matching a reference’s discretisation error is not the same as being correct, and the two should not be conflated.

4.2 Results

For the Aerosonde at $V_a^* = 25$ m/s, in the ordering of (2):

$$A_{\text{lon}} = \begin{bmatrix} -0.2821 & 0.4949 & -1.2123 & -9.7979 & 0 \\ -0.5611 & -4.4978 & 24.3714 & -0.4874 & 0 \\ 0.1986 & -3.9930 & -5.2947 & 0 & 0 \\ 0 & 0 & 1.0000 & 0 & 0 \\ 0.0497 & -0.9988 & 0 & 25.0000 & 0 \end{bmatrix}, \quad B_{\text{lon}} = \begin{bmatrix} -0.1392 & 9.4800 \\ -2.5861 & 0 \\ -36.1124 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}. \quad (19)$$

For the canard:

$$A_{\text{lon}} = \begin{bmatrix} -0.1480 & 0.4194 & 0.1624 & -9.8138 & 0 \\ -0.8219 & -5.5224 & 24.3736 & 0.0654 & 0 \\ -0.0150 & -2.2541 & -3.7468 & 0 & 0 \\ 0 & 0 & 1.0000 & 0 & 0 \\ -0.0067 & -1.0000 & 0 & 25.0000 & 0 \end{bmatrix}, \quad B_{\text{lon}} = \begin{bmatrix} -1.9051 & 9.2900 \\ -13.0192 & 0 \\ -221.0583 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}. \quad (20)$$

4.3 Structural observations

Three features of (19) are worth drawing out, because each is a usable check on the implementation.

The altitude column is identically zero. Altitude feeds back into nothing: no force or moment depends on h in a constant-density atmosphere. The system therefore has a pole exactly at the origin, altitude behaves as a pure integrator driven by the other states, and all of the interesting dynamics live in the leading 4×4 block. This is why the modal analysis of Section 6 works on that block.

The fourth row is exactly $[0 \ 0 \ 1 \ 0 \ 0]$. This is $\dot{\theta} = q$ from (6), recovered numerically. In the reference implementation the corresponding entry is 0.99997406 rather than unity, because its forward difference with a step of 0.01 cannot represent an exact identity.

The third row agrees with the analytic coefficients. The numerical Jacobian and the closed-form transfer function coefficients of Section 5 are two independent derivations of the same quantities, and they agree exactly: $A_{\text{lon}}(3, 3) = -a_{\theta_1}$ and $B_{\text{lon}}(3, 1) = a_{\theta_3}$.

5 Transfer function models

Section 5.4 of [1] develops five longitudinal transfer functions. In terms of the coefficients of Table 2,

$$\frac{\theta(s)}{\delta_e(s)} = \frac{a_{\theta_3}}{s^2 + a_{\theta_1}s + a_{\theta_2}}, \quad (21)$$

$$\frac{h(s)}{\theta(s)} = \frac{V_a}{s}, \quad \frac{h(s)}{V_a(s)} = \frac{\theta}{s}, \quad (22)$$

$$\frac{V_a(s)}{\delta_t(s)} = \frac{a_{V_2}}{s + a_{V_1}}, \quad \frac{V_a(s)}{\theta(s)} = \frac{-a_{V_3}}{s + a_{V_1}}. \quad (23)$$

The airspeed coefficient a_{V_1} is written here in a form valid for either propulsion model. The book gives the propulsive contribution for its legacy propeller as $\rho S_{\text{prop}} C_{\text{prop}} V_a^* / m$, which is exactly $-(\partial T / \partial V_a) / m$; expressing it as the derivative makes the formula model-independent:

$$a_{V_1} = \frac{\rho V_a^* S}{m} \left(C_{D_0} + C_{D_\alpha} \alpha^* + C_{D_{\delta_e}} \delta_e^* \right) - \frac{1}{m} \frac{\partial T}{\partial V_a}, \quad a_{V_2} = \frac{1}{m} \frac{\partial T}{\partial \delta_t}. \quad (24)$$

Transfer functions are returned as numerator and denominator coefficient vectors rather than as `tf` objects, so that no Control System Toolbox licence is required anywhere in the chain.

6 Reduced-order modes

Section 5.6 of [1] derives two classical approximations by holding one state constant in the fourth-order longitudinal block. The short-period approximation holds u fixed and retains (α, q) ; the phugoid approximation holds α fixed and retains (u, θ) . Both are compared against the eigenvalues of the exact 4×4 block in Table 3 and Figure 1.

Coefficient	Aerosonde	Canard	Published ref.
a_{θ_1}	5.294738	3.746763	5.294738
a_{θ_2}	99.947422	56.352608	99.947424
a_{θ_3}	-36.112390	-221.058346	-36.112390
a_{V_1}	0.296098	0.145560	—
a_{V_2}	9.480025	9.290003	—
a_{V_3}	9.810000	9.814000	9.810000
$\partial T / \partial V_a$	-2.4325	-0.0759	-2.3522
$\partial T / \partial \delta_t$	104.2803	22.2960	90.2794

Table 2: Transfer function coefficients at $V_a^* = 25$ m/s. The reference column is from the published values of the companion code, which exist for the Aerosonde only. Reference entries are blank for a_{V_1} and a_{V_2} because those depend on the trim point, and the published trim point is not an equilibrium; they are compared separately in Section 8.

Airframe	Mode	ω_n [rad/s]	ζ	T [s]
aerosonde	Short period (exact)	11.0168	0.4443	0.637
	Short period (approx.)	11.0059	0.4449	0.637
	Phugoid (exact)	0.4988	0.2858	13.144
	Phugoid (approx.)	0.4752	0.3262	13.986
canard	Short period (exact)	8.7103	0.5328	0.852
	Short period (approx.)	8.6966	0.5329	0.854
	Phugoid (exact)	0.4788	0.1407	13.256
	Phugoid (approx.)	0.5752	0.1239	11.009

Table 3: Short-period and phugoid modes, exact eigenvalues against the section 5.6 approximations.

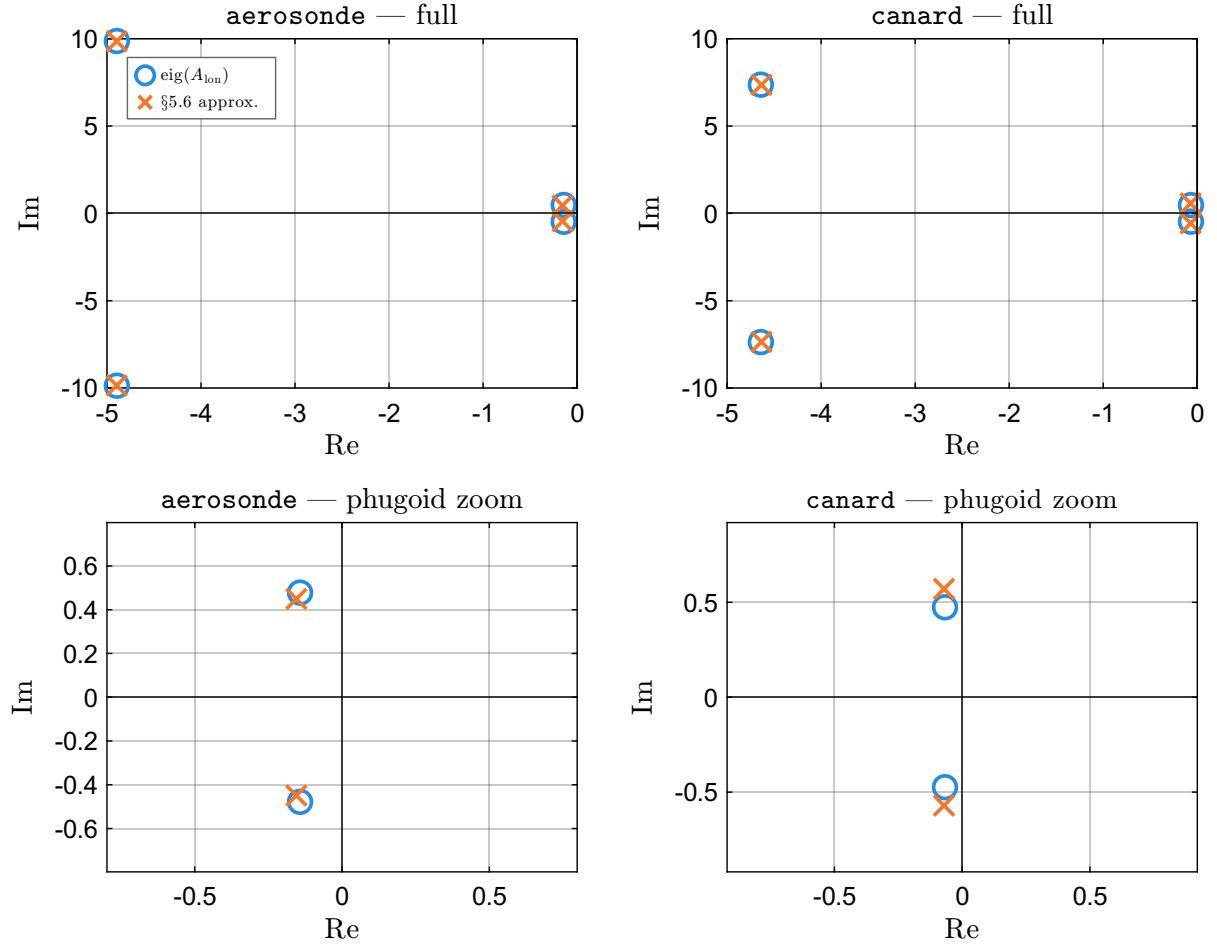


Figure 1: Longitudinal poles. Circles are eigenvalues of the leading 4×4 block of A_{lon} ; crosses are the section 5.6 approximations. The lower row zooms on the phugoid pair, which is otherwise indistinguishable from the origin at the scale set by the short-period poles.

Discussion. The short-period approximation is essentially exact for both airframes: 11.0059 against 11.0168 rad/s for the Aerosonde, an error of 0.1%, and 8.6966 against 8.7103 for the canard. This is expected. The short period is fast enough that airspeed genuinely does not change over one cycle, so holding u constant is a very good assumption.

The phugoid approximation is markedly worse, and in different directions for the two airframes: it underestimates the Aerosonde frequency by 4.7% and overestimates the canard by 20%, while getting the damping wrong by 14% and 12% respectively. This is also expected, and it is the instructive result. The phugoid is a slow exchange between kinetic and potential energy in which the angle of attack does vary, so holding α constant discards a real part of the mechanism. The approximation also neglects M_u and M_w entirely, which the exact eigenvalues retain.

The practical conclusion is that the short-period approximation is safe to design against, whereas the phugoid approximation should be treated as an order-of-magnitude estimate and any phugoid-sensitive design verified against the eigenvalues of A_{lon} .

7 Longitudinal autopilot

7.1 Structure

Section 6.4 of [1] builds the longitudinal autopilot from four loops, designed inner to outer so that each outer loop sees its inner loop as a quasi-static gain:

1. **Pitch attitude hold on elevator** (§6.4.1). Proportional-derivative, with the rate term taken from the gyro signal q directly rather than by differentiating θ : $\delta_e = k_{p\theta}(\theta^c - \theta) - k_{d\theta}q$.
2. **Altitude hold on commanded pitch** (§6.4.2). Proportional-integral, outputting θ^c .
3. **Airspeed hold on commanded pitch** (§6.4.3). Used in the climb and descend regimes.
4. **Airspeed hold on throttle** (§6.4.4). Used in the altitude-hold regime.

The proportional gain of the inner loop is set so that the elevator just saturates at the maximum anticipated pitch error, which fixes the pitch bandwidth through book equation (6.21):

$$k_{p\theta} = \frac{\delta_e^{\max}}{e_{\theta}^{\max}} \text{sign}(a_{\theta_3}), \quad \omega_{n\theta} = \sqrt{a_{\theta_2} + \frac{\delta_e^{\max}}{e_{\theta}^{\max}} |a_{\theta_3}|}. \quad (25)$$

The sign function in (25) is required for stability, not cosmetic: a_{θ_3} follows $C_{m_{\delta_e}}$, which is negative for both airframes.

Because no integral term is used on the pitch loop, its DC gain is not unity:

$$K_{\theta_{DC}} = \frac{k_{p\theta} a_{\theta_3}}{a_{\theta_2} + k_{p\theta} a_{\theta_3}}, \quad (26)$$

which evaluates to 0.619 for the Aerosonde. The outer loops must and do account for this; omitting it is a common source of steady-state altitude error.

The outer loops are separated in bandwidth from the pitch loop by factors W_h and W_{V_2} , both set to 10. A warning is raised if the resulting outer bandwidths are not in fact below $\omega_{n\theta}$, since successive loop closure is invalid otherwise.

7.2 Altitude state machine

Following book figure 6.14, the commanded pitch and throttle depend on altitude error:

Regime	Throttle	Pitch command
Take-off	$\delta_t = 1$	fixed θ^c
Climb	$\delta_t = 1$	from airspeed error
Altitude hold	from airspeed error	from altitude error
Descend	$\delta_t = 0$	from airspeed error

Airspeed is regulated with pitch rather than throttle in the climb and descend regimes, which is what keeps the aircraft away from stall when the throttle is already saturated. It is deliberately *not* used immediately after take-off, where pitching down to gain airspeed would fly the aircraft into the ground.

7.3 Implementation notes

Two implementation points proved to matter.

Integrator state is carried in an explicit struct passed in and out of the controller, never in **persistent** variables. The verification script runs the control loop more than once per session, and persistent state would make every run after the first silently wrong.

Each integrator uses conditional integration: the update is accepted only if the resulting command lies inside its limit, so the integral cannot wind up against a saturated actuator.

7.4 Results

Loop	Gain	Aerosonde	Canard
Pitch (§6.4.1)	k_{p_θ}	-4.5000	-4.5000
	k_{d_θ}	-0.4877	-0.1904
	$K_{\theta_{DC}}$	0.6192	0.9464
	ω_{n_θ}	16.2004	32.4209
Altitude (§6.4.2)	k_{p_h}	0.1884	0.2467
	k_{i_h}	0.1695	0.4443
V_a on pitch (§6.4.3)	$k_{p_{V_2}}$	-0.4313	-0.6127
	$k_{i_{V_2}}$	-0.4321	-1.1317
V_a on throttle (§6.4.4)	k_{p_V}	0.0637	0.0812
	k_{i_V}	0.0264	0.0269

Table 4: Autopilot gains from the section 6.4 formulas, with $\zeta_\theta = 0.707$, $W_h = W_{V_2} = 10$, $\zeta_h = \zeta_{V_2} = 0.9$, $\omega_{n_V} = 0.5$.

Figure 2 shows the two modes of Section 6 separating cleanly in the time domain, which is an independent confirmation of the modal analysis. Figure 3 shows both airframes tracking a 50 m altitude step to zero steady-state error while holding airspeed, with the pitch command visibly saturating during the initial climb and recovering without integrator windup.

The oscillation visible between roughly 5 and 25 s is not a numerical artefact and is worth naming. It is the altitude state machine chattering as the aircraft crosses the boundary of the altitude-hold band: within one band-width of the commanded altitude the controller switches between the climb law, which commands full throttle and regulates airspeed with pitch, and the hold law, which regulates altitude with pitch and airspeed with throttle. The two laws demand

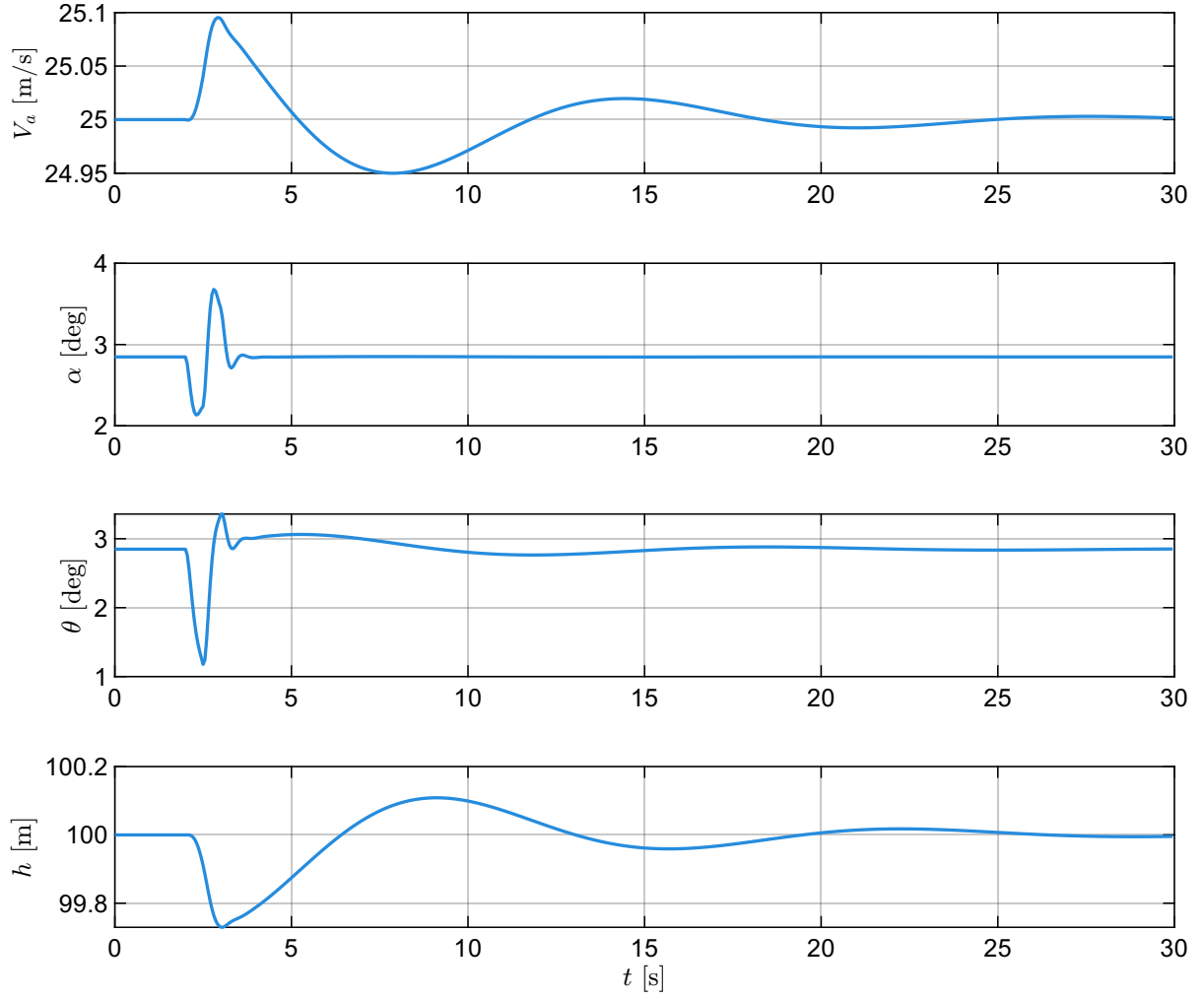


Figure 2: Open-loop response of the Aerosonde to a $\pm 2^\circ$ elevator doublet applied at $t = 2$ s, starting from trim. The fast initial transient is the short period; the slow lightly damped oscillation in airspeed and altitude that persists for the rest of the record is the phugoid.

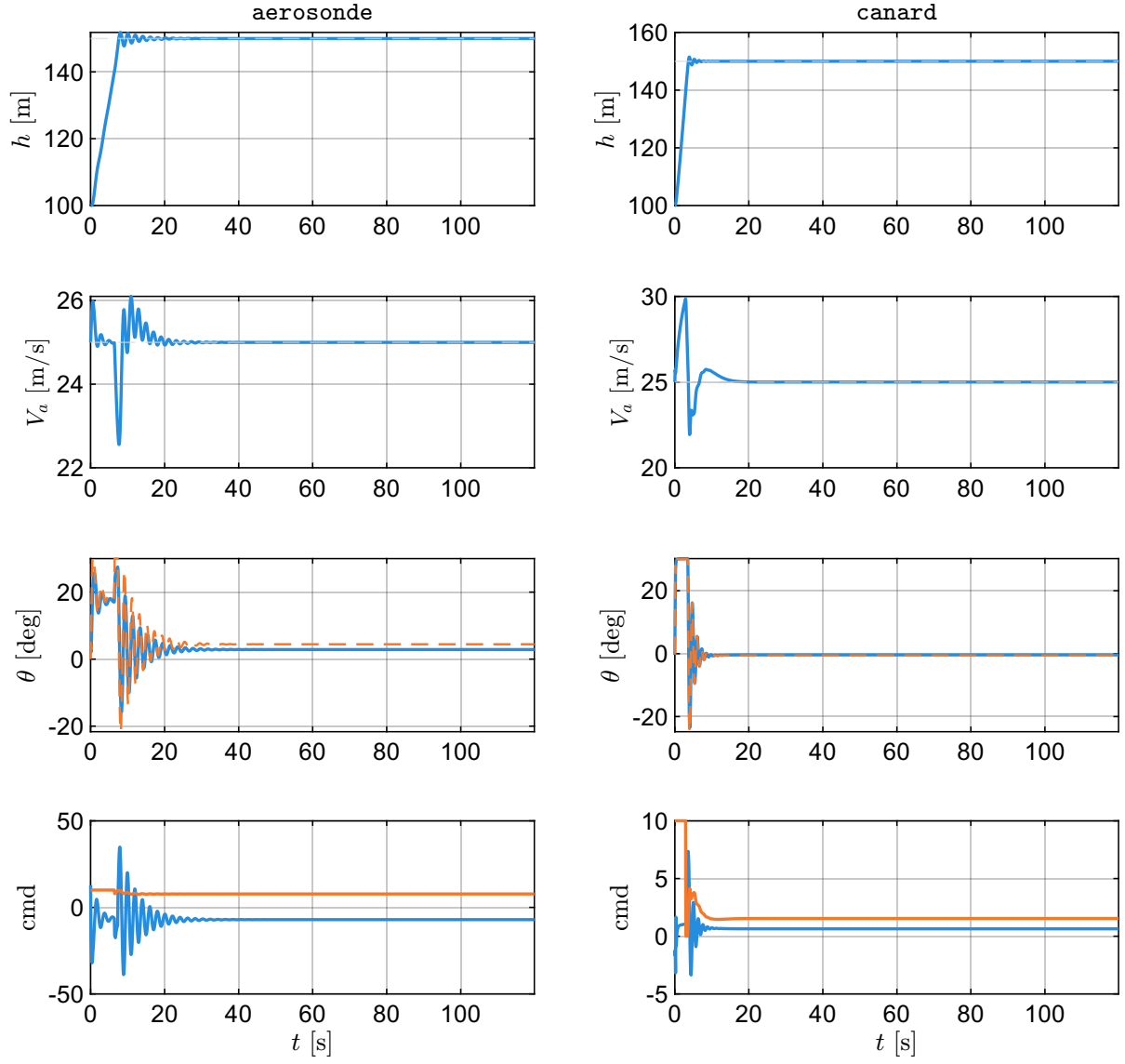


Figure 3: Closed-loop response to a +50m altitude step, both airframes. Dashed lines are commands. The bottom row shows elevator in degrees and throttle scaled by ten. Both converge to the commanded altitude with airspeed returned to its set point.

different pitch attitudes at the switching point, so repeated crossings produce the ringing. It decays because each crossing is closer to the command, and the final tracking error is zero.

This is a known limitation of the switched structure of book figure 6.14 rather than of the gain design. The standard remedies are hysteresis on the zone boundaries, or blending the two pitch commands across a transition band instead of switching discretely. Neither is implemented here, since the chapter 6 structure was followed as published.

8 Verification

Verification is organised into four groups and automated as a single script that asserts and throws. All 37 assertions pass. Two of the four groups are reference-free, which matters because the canard has no published values against which it could otherwise be checked.

Group 1: trim is a true equilibrium. $\|f(x^*, u^*)\| < 10^{-9}$ for both airframes at $\gamma = 0$ and $\gamma = 0.1$; achieved residuals are $\sim 2 \times 10^{-15}$. The climb rate is separately required to satisfy (18). Including the climbing case is what gives this group its power: a sign error in the gravity or θ terms cancels at $\gamma = 0$ and would otherwise go undetected.

Group 2: two independent derivations agree. The numerical Jacobian is checked against the closed-form section 5.4 coefficients. $A_{\text{lon}}(3, 3) = -a_{\theta_1}$ and $B_{\text{lon}}(3, 1) = a_{\theta_3}$ hold exactly for both airframes, and $A_{\text{lon}}(4, 3) = 1$ and $A_{\text{lon}}(:, 5) = 0$ hold to 10^{-9} . Agreement between two independent derivations is a stronger statement than either matching a stored number.

Group 3: against the published reference. Table 5 summarises. The agreement on A_{lon} and B_{lon} is the substantive result: differencing the same way the reference did reproduces its matrices to within 4×10^{-5} and 1.4×10^{-10} respectively.

Quantity	Agreement	Evaluated at
$a_{\theta_1}, a_{\theta_2}, a_{\theta_3}, a_{V_3}$	exact to 6–7 significant figures	own trim
$a_{V_1}, \partial T / \partial V_a$	$\sim 1 \times 10^{-4}$ relative	reference trim point
$a_{V_2}, \partial T / \partial \delta_t$	8.5×10^{-3} relative	reference trim point
A_{lon} (forward difference, step 0.01)	3.8×10^{-5} scaled	reference trim point
B_{lon} (forward difference, step 0.01)	1.4×10^{-10} scaled	reference trim point

Table 5: Comparison against published reference values. Quantities that depend on the trim point are evaluated at the reference’s own point, for the reason given in Section 9.

The looser agreement on a_{V_2} and $\partial T / \partial \delta_t$ is attributable to differencing step rather than to a modelling difference. Thrust is strongly curved in throttle near cruise, so a coarse forward difference and an accurate central difference genuinely disagree at the percent level. A hand calculation with a forward step of 0.01 gives 90.69 against the reference 90.279, while the converged central-difference value is 89.51; the reference lies between the two, consistent with its having been produced by a moderate forward step.

Group 4: closed loop. A 10 m altitude step must settle to within 0.5 m with airspeed held to 2 m/s, the pitch command inside its limit and the throttle inside $[0, 1]$, for both airframes. This group is what catches a sign error in (25) or an inverted loop ordering, both of which produce a model that trims perfectly and then diverges.

One subtlety surfaced here and is worth recording, because the first version of this check was wrong. The limit θ^{max} applies to the pitch *command*, not to the pitch *state*. A second-order

loop with $\zeta = 0.707$ overshoots its command, so $|\theta|$ legitimately exceeds $\theta^{\max}$ transiently; the canard peaks at 30.09° against a 30° limit. The assertion is therefore tight on θ^c and uses a 25% band on θ purely as a divergence guard.

9 Defects in the reference material

Three findings about the reference material are recorded here, because each changed how this work was carried out and the first two would mislead anyone repeating it.

9.1 The companion code contains no implemented physics

The official code companion to [1] is a teaching skeleton with every physics body removed, not a reference implementation. Its trim routine reads, in full relevant part,

```
state0 =
delta0 =
x0 = [ state0; delta0 ];
```

which is not syntactically valid MATLAB and will not parse. Its state-space routine is six function headers with six empty bodies; its force-and-moment function assigns its output from two variables that are never defined. The repository README confirms that worked solutions are distributed to instructors only.

Consequently nothing in this work is ported from it. All equations are taken from the text. The companion material contributed exactly two usable things: the Aerosonde parameter values, and a file of published numerical results used here as a verification oracle.

9.2 The published trim point is not a force equilibrium

Evaluating the dynamics at the published x^* and u^* gives $\dot{u} = -0.85 \text{ m/s}^2$, not zero. The cause is traceable: at $\delta_t = 0.6768$ and $V_a = 25 \text{ m/s}$ the propeller model of (16) delivers about 0.95 N of thrust, whereas the drag at that condition is 10.3 N . The equilibrium throttle is 0.7734 , which is what the trim routine here returns. This is consistent with the companion repository's own README, which notes parameter difficulties and recommends $V_a = 35 \text{ m/s}$ for this airframe.

The published A_{lon} and B_{lon} remain internally consistent, because a Jacobian and the analytic coefficients are both defined at a point and neither requires that point to be an equilibrium. This is precisely why a_{θ_1} through a_{θ_3} matched immediately while a_{V_1} and a_{V_2} did not: the former depend only on V_a and the airframe, the latter on the trim point.

The verification strategy follows from this. Trim is not compared against the published point at all. Instead the Jacobian is evaluated *at* the published point for an exact comparison of A_{lon} and B_{lon} , while the trim routine is separately required to produce a genuine equilibrium. Both checks are then tight, rather than one being confounded by the other.

9.3 The shipped parameter sets disagree with each other

The companion repository ships two copies of the same aircraft whose drag parameters differ materially:

	MATLAB copy	Python copy
C_{D_0}	0.043	0.0424
C_{D_α}	0.030	0.132
C_{D_p}	0.0	0.043
c	0.19	0.18994

C_{D_α} differs by a factor of 4.4, which is not a rounding difference. The published reference values were generated with the Python set. This was settled by direct computation rather than by inspection: substituting into (24),

$$a_{V_1} = \frac{1.2682 \times 25 \times 0.55}{11} (0.0424 + 0.132(0.05001) + 0.0135(-0.12478)) + \frac{2.3522}{11} = 0.288845, \quad (27)$$

against the published 0.2888454, agreeing to seven significant figures. The MATLAB values do not reproduce it. The parameter file used here therefore carries the Python numbers, with an explicit warning comment against “correcting” them back.

Equation (27) also settles two secondary questions at once. It confirms the linear-in- α drag law of (15) rather than the induced-drag polar, and the magnitude of its propulsive term confirms the addendum propeller model rather than the legacy one.

10 Conclusions

The longitudinal design chain of [1] chapters 3 to 6 has been implemented in full and verified. The principal results are:

- Trim is solved to a residual of $\sim 2 \times 10^{-15}$ for both airframes, at level and climbing conditions, using two scalar root-finds rather than a constrained optimiser. Exploiting the triangular structure of the problem removes the Optimization Toolbox dependency and improves the residual by nine orders of magnitude over a tolerance-based solve.
- The linearised model reproduces published reference matrices to 3.8×10^{-5} and 1.4×10^{-10} , and the analytic transfer function coefficients to seven significant figures.
- The short-period approximation of section 5.6 is accurate to 0.1% and safe to design against. The phugoid approximation errs by 5 to 20% in frequency and should be treated as indicative only.
- The successive-loop-closure autopilot tracks altitude steps to zero steady-state error on both airframes while holding airspeed, with the inner-loop DC gain of (26) correctly propagated to the outer loops.

Two defects in the reference material were identified and documented: a companion codebase with no implemented physics, and a published trim point that is not a force equilibrium. The second of these required restructuring the verification so that reference comparison and equilibrium verification are separate, independently tight checks rather than a single confounded one.

The natural extensions are the lateral chain, which would reuse the trim and linearisation machinery essentially unchanged, and atmospheric disturbance, which enters as an additive term on V_a and α and would leave the rest of the structure intact.

A Software

Eleven MATLAB files, roughly 1300 lines, with no toolbox dependencies.

File	Chapter	Purpose
<code>lon_params.m</code>	—	Both airframes, one canonical struct
<code>f_lon.m</code>	3, 4	Nonlinear EOM, forces, both thrust models
<code>compute_lon_trim.m</code>	5.3, F.2	Trim
<code>compute_lon_ss.m</code>	5.5, F.4	Finite-difference Jacobian
<code>compute_lon_tf.m</code>	5.4	Transfer functions and coefficients
<code>lon_modes.m</code>	5.6	Modal analysis, exact and approximate
<code>lon_gains.m</code>	6.4	Gain design
<code>lon_autopilot.m</code>	6.4	Four loops and the state machine
<code>lon_autopilot_reset.m</code>	6.4	Integrator state
<code>run_lon.m</code>	—	Driver
<code>check_lon.m</code>	—	37 verification assertions

Reproduce the results of this report with

```
run_lon
check_lon
```

and regenerate its tables and figures with `make_report_data` from the report directory.

References

- [1] R. W. Beard and T. W. McLain, *Small Unmanned Aircraft: Theory and Practice*, Princeton University Press, 2012.